

PROJECT 01

RCM Claims & Revenue Cycle Analysis

Excel-based healthcare revenue cycle analytics case study

Domain: US Healthcare Revenue Cycle Management (RCM) | Tool: Microsoft Excel | Dataset: Synthetic, 3,000 claims

1. Executive Summary

This project analyzes a synthetic healthcare claims dataset to identify where financial exposure is concentrated across Accounts Receivable (AR), claim denials, payer/category combinations, and monthly paid revenue. The analysis was designed around five business questions and translated the results into an interactive Excel dashboard.

The analysis found that the 120+ day AR bucket contains 32.46% of total outstanding AR. Coding, Authorization, Eligibility, and Medical Necessity account for 76.63% of total denial dollars, while adding Documentation raises cumulative coverage to 90.81%.

The Top 10 payer × denial combinations represent \$212,052.23, or 43.41% of total denial dollars. Paid revenue peaked in April at \$616,495.78, fell to \$497,701.63 in May, recovered in June, and declined slightly in July.

This project identifies financial concentration and investigation priorities; it does not claim that the patterns prove root causes or guaranteed recoverability.

2. Business Context

Healthcare providers submit claims to payers for services delivered to patients. Unpaid balances become Accounts Receivable, while claims or claim amounts that are not paid because of an issue can become denials.

Revenue Cycle Management (RCM) teams work across the claim lifecycle to support accurate billing, payment collection, denial resolution, appeals, and follow-up of outstanding receivables.

Project objective: convert claim-level operational data into a management-friendly view that identifies financial concentration, trends, and priority areas for investigation.

3. RCM Concepts for Non-RCM Readers

Claim — A request submitted to a payer for payment for healthcare services.

Payer — The organization responsible for paying a healthcare claim.

Allowed Amount — The amount recognized as payable under the applicable payer/contract terms.

Paid Amount — The amount actually paid against the claim.

Accounts Receivable (AR) — Money that has been billed but has not yet been collected.

AR Aging — Grouping outstanding AR by how long it has remained unpaid.

Denial — A claim or claim amount that the payer does not pay because of an identified issue or requirement.

Denial Category — A broad reason associated with a denial, such as Coding, Authorization, Eligibility, or Documentation.

Payer × Denial Category — A combination used to identify which payer and denial type are associated with a particular amount of financial exposure.

4. Business Questions

BQ01 — Where is outstanding AR concentrated?

BQ02 — Which denial categories drive financial exposure?

BQ03 — Which payer/category combinations drive denial dollars?

BQ04 — How does paid revenue change over time?

BQ05 — Which areas require prioritization?

5. Dataset & Analytical Scope

The supplied workbook contains a synthetic claim-level dataset with 3,000 records. The primary RCM_Analysis sheet contains 35 fields covering claim identifiers, dates, payer and specialty dimensions, financial amounts, AR aging, denial information, appeal information, and validation/check fields.

Core analytical fields include Claim_ID, Service_Month, Payer, Specialty, Allowed_Amount, Paid_Amount, Outstanding_AR, AR_Days, Aging_Bucket, Denial_Flag, Denial_Category, and Denial_Amount.

QA note: the older Project_Control sheet lists 29 dataset columns, but the current RCM_Analysis sheet contains 35 fields. The control-sheet metadata should be updated before publication.

6. Analytical Methodology

Data preparation: reviewed the claim-level structure and validation/check fields and prepared PivotTable-based aggregations.

Analysis: AR aging, denial concentration, payer performance, monthly trends, payer × denial analysis, Pareto analysis, and Top 10 prioritization.

Visualization: KPI cards, trend charts, payer comparisons, AR aging, Pareto analysis, payer × denial heatmap, Top 10 prioritization, and slicers for Payer, Service Month, and Specialty.

7. Findings & Interpretation

BQ01 — Where is outstanding AR concentrated?

Finding: The 120+ day aging bucket contains \$355,589.53 of outstanding AR, representing 32.46% of total outstanding AR.

RCM interpretation: A substantial portion of unpaid receivables has moved into the oldest aging bucket and warrants focused follow-up.

Plain-English interpretation: Nearly one-third of unpaid money has remained outstanding for more than 120 days.

Recommended investigation: Review 120+ AR by payer, claim status, denial status, balance, and next-action status.

BQ02 — Which denial categories drive financial exposure?

Finding: Coding, Authorization, Eligibility, and Medical Necessity account for 76.63% of total denial dollars. Adding Documentation increases cumulative coverage to 90.81%.

RCM interpretation: Denial exposure is concentrated in a relatively small number of categories.

Plain-English interpretation: Most denied dollars come from a few recurring issue types.

Recommended investigation: Prioritize Coding, Authorization, Eligibility, and Medical Necessity, with Documentation as the next significant area.

BQ03 — Which payer/category combinations drive denial dollars?

Finding: UnitedHealthcare has 60.73% of its denied dollars concentrated in Authorization, Medical Necessity, and Eligibility. Cigna's largest category is Coding, Aetna's largest category is Eligibility, and Medicare and BCBS show their largest category exposure in Coding. Medicaid shows its largest category exposure in Authorization.

Plain-English interpretation: The denial problem is not uniform across payers; the category driving exposure can change depending on the payer.

Recommended investigation: Use payer-specific workstreams for the highest-exposure payer × category combinations.

BQ04 — How does paid revenue change over time?

Finding: Paid revenue increased from January (\$564,588.19) through April (\$616,495.78), the highest observed month. It declined to \$497,701.63 in May, recovered in June, and declined slightly in July.

Important limitation: The current analysis identifies the movement but does not establish why the May decline occurred.

Recommended next analysis: Decompose monthly paid revenue by claim volume, payer mix, payment rate, denial activity, and AR movement.

BQ05 — Which areas require prioritization?

Finding: The four largest denial categories cover 76.63% of denial dollars. The Top 10 payer × denial combinations total \$212,052.23, representing 43.41% of total denial dollars.

Interpretation: WHAT to investigate = high-impact denial categories. WHERE to investigate = specific high-exposure payer × category combinations.

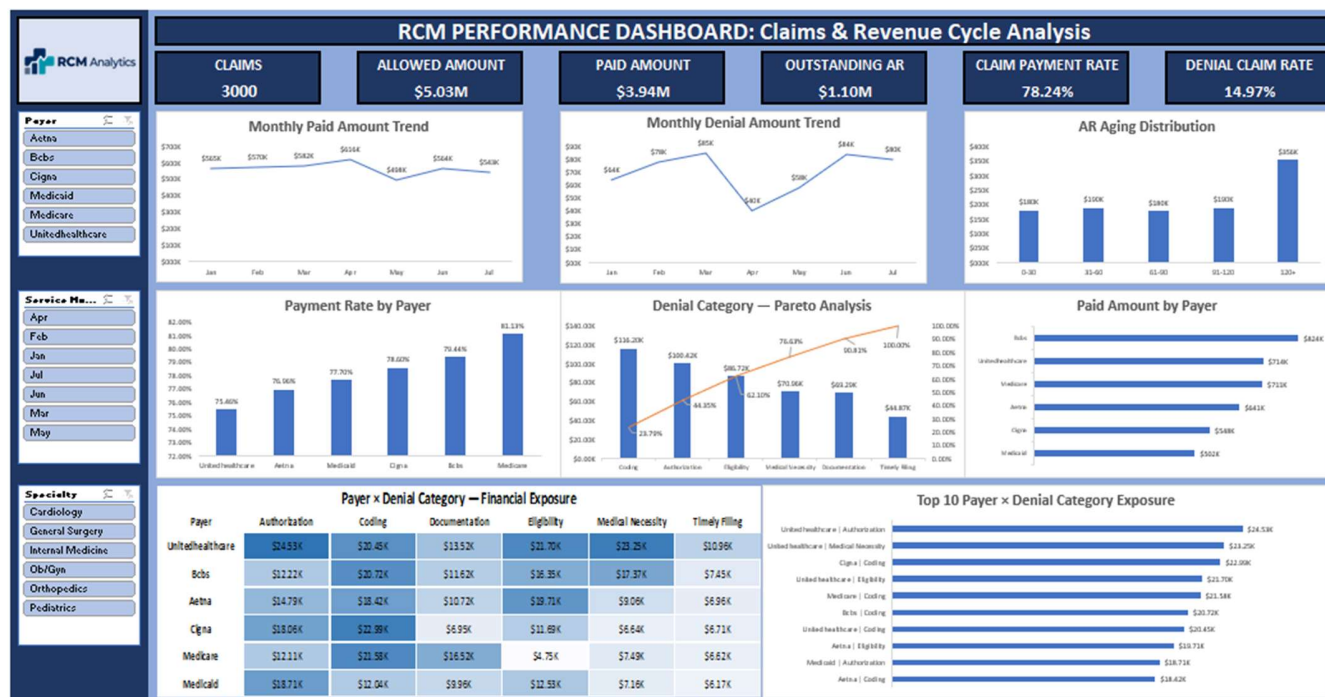
Important wording: This is prioritization for investigation, not a claim that all identified denied dollars are recoverable.

8. Executive KPI Snapshot

KPI	Value	Definition
Total Claims	3,000	Total claim records
Allowed Amount	\$5,034,573.60	Total allowed amount
Paid Amount	\$3,939,008.19	Total paid amount
Outstanding AR	\$1,095,565.41	Total unpaid AR
Payment Rate	78.24%	$\text{Paid} \div \text{Allowed}$
Denial Claim Rate	14.97%	$\text{Denied claims} \div \text{total claims}$
Total Denial Amount	\$488,466.91	Total denied dollars

9. Dashboard

The final dashboard consolidates the project into an interactive management view. It contains six executive KPIs, monthly paid and denial trends, AR aging, payer payment performance, paid amount by payer, denial-category Pareto analysis, payer × denial heatmap, and Top 10 payer × denial prioritization.



Final locked dashboard supplied with the project workbook.

10. Recommendations

Focus denial RCA on the highest-exposure categories: Coding, Authorization, Eligibility, and Medical Necessity, then expand to Documentation.

Use payer-specific denial analysis and investigate the largest payer × category combinations individually.

Establish targeted 120+ AR review segmented by payer, claim, denial status, balance, and next action.

Investigate payment volatility by decomposing monthly paid revenue into claim volume, payer mix, payment rate, denial activity, and AR movement.

Monitor the Top 10 payer × denial combinations over subsequent periods to determine whether exposure is increasing, decreasing, or persistent.

11. Business Value

The main value of the analysis is focus: 120+ AR represents 32.46% of AR, the four highest-exposure denial categories represent 76.63% of denial dollars, and the Top 10 payer × denial combinations represent 43.41% of denial dollars.

These outputs support operational review, denial-management worklists, payer-specific investigation, and future monitoring. They are not guaranteed recovery or savings estimates.

12. Limitations & Analytical Boundaries

The dataset is synthetic and intended for portfolio analysis rather than production operational decision-making.

The analysis identifies concentration and patterns but does not establish the exact root cause of each denial.

The project does not quantify guaranteed recoverable dollars or savings.

The May payment decline is observed, but its cause is not established by the current analysis.

The relationship between aged AR and denial categories is not treated as causal without additional claim-level investigation.

13. Technical Skills Demonstrated

Excel: PivotTables, PivotCharts, slicers, conditional formatting, calculated metrics, and dashboard design.

Data Analysis: data preparation, aggregation, KPI calculation, trend analysis, concentration analysis, Pareto analysis, and prioritization.

Business Analytics: translating business questions into analytical questions, selecting metrics, interpreting evidence, and developing recommendations.

Communication: explaining RCM concepts for non-RCM audiences and distinguishing observed facts from hypotheses.

14. Final Conclusion

The RCM Claims & Revenue Cycle Analysis demonstrates how claim-level operational data can be transformed into a structured management view of financial exposure. The analysis identifies concentration in aged AR, a small number of denial categories, and specific payer × denial combinations, while also showing changes in paid revenue over time.

The dashboard and evidence framework provide a defensible starting point for RCM investigation: identify the largest areas of exposure, then narrow the investigation to the payer/category combinations contributing the most financial impact. The project deliberately separates what the data shows from what would require additional investigation.

15. Final QA — Written Numbers & Metadata

Verified: Total Claims = 3,000.

Verified: Allowed Amount = \$5,034,573.60.

Verified: Paid Amount = \$3,939,008.19.

Verified: Outstanding AR = \$1,095,565.41.

Verified: Payment Rate = 78.24%.

Verified: Denial Claim Rate = 14.97%.

Verified: 120+ AR share = 32.46%.

Verified: Top 4 denial categories = 76.63%.

Verified: Top 5 denial categories = 90.81%.

Verified: Top 10 payer × denial amount = \$212,052.23.

Verified: Top 10 share of denial dollars = 43.41%.

Verified: April paid amount = \$616,495.78; May = \$497,701.63.

Metadata issue: current RCM_Analysis sheet contains 35 fields while the Project_Control sheet says 29. Update the control sheet before publication.